Sri Chaitanya IIT Academy.,India.

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A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: Sr. XL-500, Sr.S60(N&S) & LIIT | Paper -1(Adv-2019-P1-Model) | Date: 28-12-2025

Time: 09.00Am to 12.00Pm | GTA-09 | Max. Marks: 186

KEY SHEET

PHYSICS

1	**B**	2	**B**	3	**B**	4	**A**	5	**BC**	6	**AB**
7	**BD**	8	**AC**	9	**D**	10	**AC**	11	**ABC**	12	**AC**
13	**1**	14	**6**	15	**5**	16	**8**	17	**3**	18	**1**

CHEMISTRY

19	**B**	20	**A**	21	**A**	22	**D**	23	**BD**	24	**ACD**
25	**ABCD**	26	**BCD**	27	**AC**	28	**AC**	29	**ABD**	30	**AC**
31	**20**	32	**9**	33	**6**	34	**18**	35	**64**	36	**1**

MATHEMATICS

37	**A**	38	**A**	39	**C**	40	**B**	41	**ACD**	42	**ABD**
43	**AC**	44	**BD**	45	**B**	46	**ABCD**	47	**AC**	48	**ABD**
49	**9**	50	**712**	51	**67**	52	**216**	53	**2**	54	**519**

For More Join: @IITJEE_Advanced

SOLUTIONS

PHYSICS

01. $v_p = -vx$ slope

$x = \pi/8$ at $t = \pi/6$

Particle is at mean position

02. $\%\ error = 100\left(\frac{\Delta l}{l} + \frac{4\Delta r}{r}\right) = \left(\frac{0.1}{10} + 4\left(\frac{0.01}{2}\right)\right)\times 100 = 3$

03. $\frac{(\mu-1)t}{\lambda} = 7$

04. Let $\phi_1 = 4eV$, then $\phi_2 = 2eV$

$(E-\phi)$ represent kinetic energy of most energetic electron.

$E - \phi_2 = 2(E-\phi_1) \Rightarrow E = 6eV$

05. CONCEPTUAL

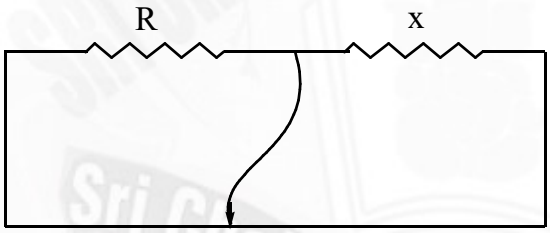
06. (C) $\frac{M}{M_g} = \frac{Mc^2}{M_0c^2} = \frac{K + M_0c^2}{M_0c^2} = 1 + \frac{10^{-9}}{1.67\times10^{-27}\times9\times10^{20}} \approx 7.68$

(D) $Mc^2 = M_0c^2 + K = 2m_0c^2$

$\Rightarrow M = 2m_0 \qquad \Rightarrow \frac{M_0}{\sqrt{1-\frac{v^2}{c^2}}} = 2M_0$

$\Rightarrow 1 = 4 - \frac{4v^2}{c^2} \qquad \Rightarrow v^2 = \frac{3c^2}{4} \Rightarrow v = \frac{\sqrt{3}}{2}c = 0.866c$

07.



$\frac{R_1}{x} = \frac{40}{60} = -\frac{2}{3} \qquad \Rightarrow R_1 = \frac{2}{3}x$

Now, $\frac{R_2}{x} = \frac{50}{50} \Rightarrow R_2 = x$

08. CA process is adiabatic

$3P_0V_1 = 2P_0V_0$

$V_1 = \frac{2V_0}{3}$

$PV^r = \text{constant}$

$3P_0V_1^r = P_0V_0^r$

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Work done $= nRTln\left(\frac{V_2}{V_1}\right)$

09. $J = \sigma E \Rightarrow \frac{I}{4\pi r^2} = S\left(\frac{Q_1 + Q_2 - Q}{4\pi\varepsilon_0 K r^2}\right)$

$Q = \frac{3}{25}\left(75^2 - 30^2 - 40^2\right) + 2(75 - 30 - 40)$

$Q > 0$

Similarly $A_{10} + A_{20} \longrightarrow A_{30} + \theta$, releases energy and $A_{80} \rightarrow 2A_{40}$; do not releases energy.

For $A > 100$

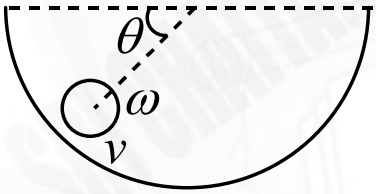
$E = 12A - \frac{A^2}{25}$ is the binding E of a nuclear

$A_{150} \longrightarrow 2A_{75} + Q$

$Q = [2(8)(75)] - \left[12(150) - \frac{(150)^2}{25}\right]$

$Q > 0$

10. Magnetic forces due to angular motion of sphere will not create any torque. Magnetic forces due to translational motion passes through centre of mass of sphere acts parallel to Normal reaction



$v = r\omega$ .....(1)

$\frac{1}{2}\left(\frac{2}{5} + 1\right)mr^2\left(\frac{v}{r}\right)^2 = mg(R - r)\sin\theta$

$v = \sqrt{\frac{10}{7}g(R - r)\sin\theta}$ .....(2)

$m\frac{v^2}{(R - r)} = N - mg\sin\theta - qvB$

In limiting condition $N = 0; \theta = 90^\circ$

$qvB = mg + \frac{10}{7}mg = \frac{17mg}{7}$

$B_{\text{limiting}} = \frac{17mg}{7qv} = 2T$

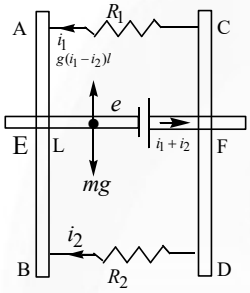
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11. $$\frac{n_P}{n_P\left(\frac{d}{n_P}+\frac{t}{1}+\frac{x}{n_G}\right)}+\frac{1}{\infty}=\frac{(1-n_P)}{-R}$$

$$\Rightarrow \frac{d}{n_P}+t=\frac{R}{n_P-1}-\frac{x}{n_G}$$

12. $m\vec{v}_B+2m\vec{v}_{AC}=0 \quad \vec{v}_{AC}=\frac{-\vec{v}_B}{2}$

13. We can understand the direction of flow of induced currents if a fictitious battery is attached between EF. The direction of induced current can be found with the help of Lenz's law.



[P.d. across parallel combination remains the same]

Also, $P_1=ei_1=0.76W$ and $P_2=ei_2=1.2W$.

$$\therefore \frac{i_1}{i_2}=\frac{0.76}{1.2}\Rightarrow i_1=\frac{0.76}{1.2}i_2 \ldots.(ii)$$

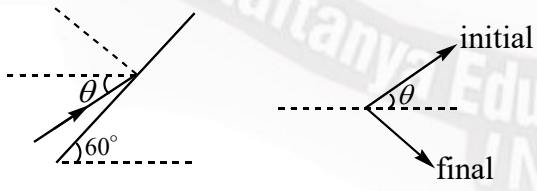
14. Suppose the liquid rises to a height h. Then the total energy of the liquid in the capillary is

$$E(h)=\frac{\pi}{4}\left(d_2^2-d_1^2\right)h\times\rho g\times\frac{h}{2}-\pi\left(d_2-d_1\right)\alpha h$$

Minimizing E we get $h=\frac{4\alpha}{\rho g\left(d_2-d_1\right)}=6cm$

15. $r_1=r_2=30^\circ \quad 1(\sin i_1)=\sqrt{2}\left(\frac{1}{2}\right)=\frac{1}{\sqrt{2}}$

$i_1=45^\circ \qquad \theta=15^\circ$



Charge in momentum of each photon $=2P\sin(15^\circ)$

$$\text{Force}=\left(\frac{P_0}{P_c}\right)2P\sin(15^\circ)=\frac{2P_0}{c}\sin 15=\frac{P_0}{c}\left(\frac{\sqrt{3}-1}{\sqrt{2}}\right)$$

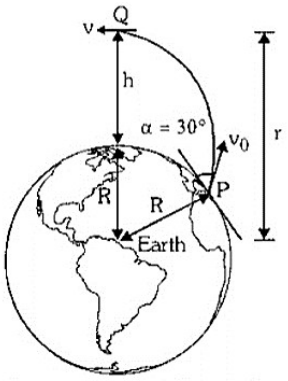
$x=3 \quad y=2$

16. Let velocity at highest point be v and R + h = r

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Applying conservation of angular momentum between P and Q, we have

$mvr = mv_0\ R\cos 30^\circ$ Or $v = \dfrac{\sqrt{3}v_0 R}{2r}$



Applying conservation of mechanical energy between P and Q, we have

$\dfrac{1}{2}mv_0^2 - \dfrac{GMm}{R} = \dfrac{1}{2}mv^2 - \dfrac{GMm}{r}$

Substituting the value of v from equation (1), we get

$\dfrac{1}{2}mv_0^2 - \dfrac{GMm}{R} = \dfrac{1}{2}m\left\{\dfrac{3v_0^2R^2}{4r^2}\right\} - \dfrac{GMm}{r}$ Or $v_0^2 - \dfrac{2GM}{R} = \dfrac{3v_0^2R^2}{4r^2} - \dfrac{2GM}{r}$

Or $\dfrac{1.5GM}{R} - \dfrac{2GM}{R} - \dfrac{3}{4}\left(\dfrac{1.5GM}{R}\right)\dfrac{R^2}{r^2} - \dfrac{2GM}{r}$

$\left(v_0 = \sqrt{\dfrac{1.5GM}{R}}\ is\ given\right)$ Or $-\dfrac{1}{2R} = \dfrac{9}{8R}.\dfrac{R^2}{r^2} - \dfrac{2}{r}$

Or $-4r^2 - 9R^2 - 16Rt$ Or $4r^2 - 16Rr + 9R^2 = 0$

Or $r = \dfrac{16R \pm \sqrt{256R^2 - 144R^2}}{8}$

$\Rightarrow = \dfrac{16R \pm 10.58R}{8} = 3.323R\ and\ 0.677R$

But r < R

Hence r = 3.323 R

Or h = r – R = 2.323 R

17. Using condition of equilibrium

$KL = (m_1 + m_2)g$

Using COME

$\dfrac{1}{2}KL^2 = m_1 g(2L)$

$\Rightarrow m_1 = \dfrac{KL}{4g} = \dfrac{(m_1 + m_2)}{4}$

$\Rightarrow 3m_1 = m_2 = m_2 / m_1 = 3$

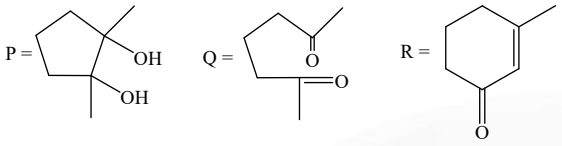
18. Use dimensional analysis

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CHEMISTRY

19. $v) \rightarrow$ No reaction

20.



21. $rate = K[N_2O_2][H_2] \qquad K_{eq} = \dfrac{[N_2O_2]}{[NO]^2}$

$N_2O_2 = K_{eq} = [NO]^2$

$rate = K.K_{eq}[NO]^2.[H_2]$

$rate = 2.1\times10^{-7}\ moles\ l^{-1}\min t^{-1}$

22. $XeF_2 : sp^3d$:linear (planar)

$XeF_4 : sp^3d^2$: square planar

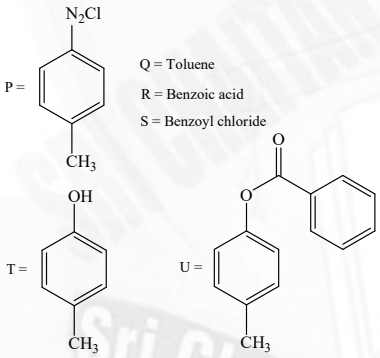
$ICl_4^- : sp^3d^2$: square planar

$SO_3 : sp^2$:Trigonal planar

23. $A) \rightarrow SIP$ decreases basic strength.

$D) \rightarrow$ intra molecular hydrogen bonding decreases acidic strength.

24.



25. (i) Fe extraction: slag: $CaSiO_3$

(ii) CO act as a RA

26. CONCEPTUAL

27. Tube A contains 5 moles Cl_2

Tube B contains 10 moles PCl_3

$PCl_3 + Cl_2 \rightarrow PCl_5$

10	205	0
7.5	0	2.5

After above reaction

Tube A contains 2.5 mole Cl_2

Tube B contains 7.5 mole PCl_3 and $2.5\,mole$ PCl_5

For More Join: @IITJEE_Advanced

Now 3.75 mole PCl_3 and 1.25mole PCl_5 is transferred to Tube A

$PCl_3 + Cl_2 \rightarrow PCl_5$

3.75 2.5 1.25

1.25 0 3.75

Finally In tube A

$PCl_5 = 3.75 mole$

$PCl_3 = 1.25 mole$

In tube B

$PCl_5 = 1.25 mole$

$PCl_3 = 3.75 mole$

$For\ Ne:$

28. $C_{v_m} = \frac{3R}{2}\ C_{P_m} = \frac{5R}{2}\ Y = 1.66$

$For\ O_2:$

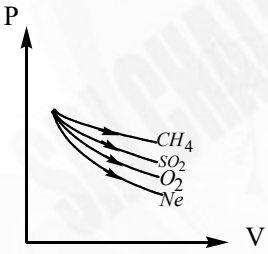
$C_{v_m} = \frac{7R}{2}\ C_{P_m} = \frac{9R}{2}\ Y = 1.29$

$For\ SO_2:$

$C_{v_m} = 6R\ \ C_{P_m} = 7R\ Y = 1.16$

$For\ NH_3:$

$C_{v_m} = 12R\ \ C_{P_m} = 13R\ Y = 1.08$



Step-II: Isochoric heating

$\Delta U = \Delta V_1 + \Delta V_2$

$\Delta U = nC_{V_m}\left[T_f - T\right] + nC_{V_m}\left[T - T_f\right] \qquad \Delta U = 0$

29. Mohr's salt: Fe^{+2} to Fe^{+3} Oxalic acid: $C_2O_4^{2-}$ to CO_2

Ferric sulphate: $Fe_2(SO_4)_3 : Fe^{+3}$

H_2O_2 : act as reducing agent with $KMnO_4$.

30. (i) $E^0_{Co^{3+}/Co^{2+}} > E^0_{Mn^{3+}/Mn^{2+}} > E^0_{Fe^{3+}/Fe^{2+}}$

(ii) $MnO_2 \rightarrow K_2\overset{+6}{M}nO_4 \xrightarrow[oxidation]{electrolytic} K\overset{+7}{M}nO_4$

31. X = 6, Y = 10, Z = 4

For More Join: @IITJEE_Advanced

32. In 294 gram gm eq of $HCl = \frac{294}{36.5} = 8.05$

gm eq of $H_2SO_4 = \frac{294}{49} = 6$ gm eq of $H_3PO_4 = \frac{294}{32.66} = 9$

In any possible case number of gm eq of acid will not be more than 9

gm eq of Acid = gm eq of $NaOh$

$9 = M \times n.f \times V(l)$

$9 = 1 \times 1 \times V(l)$

$V(l) = 9$

33. Balanced equation

$3I_2(S) + 6OH^- \longrightarrow 5I^- + IO_3^- + 3H_2O$

$\Delta G^0_{reaction} = -172.5Kg \quad \Delta G^0_{reaction} = -RT \ln k_{eq}$

$k_{eq} = 10^{30} \quad k_{eq} = \frac{\left[1O_3^-\right]\left[I^-\right]^5}{\left[0H^-\right]^6}$

$\left[OH^-\right]^6 = 10^{-36} \quad \left[OH^-\right] = 10^{-6} \quad P^{OH} = 6$

34.

$MX \longrightarrow M^+ + X^-$

$S \quad S$

$NaX \longrightarrow Na^+ + X^-$

$10^{-6} \quad 10^{-6}$

$\left[M^+\right] = S \quad \left[X^-\right] = \left(S + 10^{-6}\right)$

$\left[Na^+\right] = 10^{-6} \qquad K = K_{Na^+} + K_{M^+} + K_{X^-}$

$29 \times 10^{-6} = 5 \times 10^{-3} \times 1000 \times 10^{-6} + 6 \times 10^{-3} \times 1000 \times S$

$+4 \times 10^{-3} \times 1000\left(S + 10^{-6}\right)$

$S = 2 \times 10^{-6} \quad K_{sp} = \left[M^+\right]\left[X^-\right] = 2 \times 10^{-6} \times 3 \times 10^{-6}$

$= 6 \times 10^{-12}$

$NO + O_3 \longrightarrow NO_2 + O_2$

35. 30 48 32

60 96 ?

36. (i) Complex-I: $\left[Cu(NH_3)_4\right]SO_4$

(ii) Complex-II: $Na_2\left[Zn(OH)_4\right]$

For More Join: @IITJEE_Advanced

MATHEMATICS

37. Normal to $y^2 = 4ax$ with slope m is

$y = mx - 2am - am^3$ .....(1)

(1) touches $x^2 - y^2 = a^2$ if

$(-2am - am^3)^2 = a^2(m^2 - 1)$

i.e. if $m^6 + 4m^4 + 3m^2 + 1 = 0$ .....(2)

(2) is a sixth degree equation in m with no odd power terms and whose coefficients are all positive.

$\Rightarrow$ (2) has no real root

$\therefore |S| = 0$

38. $|AB| = 0$ as $AB = \begin{bmatrix} x+1 & x+2 & 0 \\ 3x+3 & x+5 & 0 \\ 6x+2 & x+4 & 0 \end{bmatrix}\begin{bmatrix} x+1 & x+2 & 5x+7 \\ x+2 & x+7 & 8x \\ 0 & 0 & 0 \end{bmatrix}$

39. $g(x) = mx$

Also $f^1(1) = f(-1) \Rightarrow g(x) = -\frac{2}{3}\left(\frac{1}{6} + \ln\frac{3}{2}\right)x$

40. $\frac{a\pi}{b} = \frac{13\pi}{7}$

$\Rightarrow a = 13, b = 7$

41. Differentiating with respect to x

$y.f'(xy) = e^{xy-x}.f'(x) + f(x)e^{xy-x}(y-1) + f(y).e^{xy-y}.y$

Put $x = 1$,

$\Rightarrow y.f'(y) = e^{y-1}.f'(1) + f(1)e^{y-1}(y-1) + yf(y)$

$f(1) = 0$ [$put\ x = y = 1$]

$\Rightarrow y.f'(y) = e^y + y.f(y)$

$\Rightarrow f(x)e^x.\ln x$

42. Total no. of matrices $= {}^aC_3 = 84$

No. of non-singular matrices $= 3 \times 2 \times 1 = 6$

No. of matrices with zero trace $= {}^6C_3 = 20$

43. Let tangent be $y + \frac{1}{2} = mx + \frac{P}{m}$

$x^2 = 2y$

$x^2 = 2\left(mx + \frac{P}{m} - \frac{1}{2}\right)$

$x^2 - 2mx - \frac{2P}{m} + 1 = 0$

For More Join: @IITJEE_Advanced

$\Delta = 0 \;\; 4m^2 = 4\left(1 - \frac{2P}{m}\right)$

$f(m) = m^3 - m + 2P = 0$

$f'(m) = 3m^3 - 1 = 0 \;\; m = \pm\frac{1}{\sqrt{3}}$

$f\left(\frac{1}{\sqrt{3}}\right) = 2P - \frac{2}{3\sqrt{3}}$

$f\left(-\frac{1}{\sqrt{3}}\right) = 2P + \frac{2}{3\sqrt{3}}$

$f(m)$ has one root if $f\left(\frac{1}{\sqrt{3}}\right).f\left(-\frac{1}{\sqrt{3}}\right) > 0$

$f(m)$ has 3 roots if $f\left(\frac{1}{\sqrt{3}}\right).f\left(-\frac{1}{\sqrt{3}}\right) < 0$

44. $\{P\} = 0 \;\Rightarrow P \in N \;\&\; P \geq 2$

$Q = Q^2 - 3Q \Rightarrow Q = 4$

45. $I + f = (4 + \sqrt{15})^n$

Let $g = (4 - \sqrt{15})^n$, then $0 < g < 1$

$I + f = {}^nC_0 4^n + {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 4^{n-2}15 + {}^nC_3 4^{n-3}(\sqrt{15})^3 + \ldots.$

$g = {}^nC_1 4^{n-1}\sqrt{15} + {}^nC_2 4^{n-2}15 - {}^nC_3 4^{n-3}(\sqrt{15})^3 + \ldots.$

$\therefore I + f + g = 2({}^nC_0 4^n + {}^nC_2 4^{n-2}15 + \ldots\ldots) = \text{even integer}$

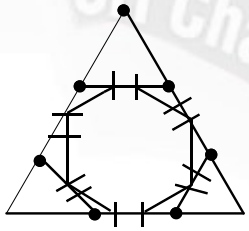
$\because \; 0 < f + g < 2$

$\Rightarrow f + g = 1$

$\Rightarrow 1 - f = g$ thus I is an odd integer $1 - f = g = (4 - \sqrt{15})^n$

$(I + f)(1 - f) = (I + f)g = 1$

46. $P_0 \qquad area = 10 \qquad side\ length = x$



Number of sides

$P_0 - 3; \qquad P_1 - 6; \qquad P_2 - 12 \ldots\ldots P_n \to 2^n \times 3$

$(P_1) area = P_0 - 3 \times each\ corner\ area\ of\ P_0$

$(P_2) area = P_1 - 6\ corner\ \Delta^{le} area\ of\ P_1$

$(P_n)_A = (P_{n-1})_A - (no.of\ corner\ of\ P_{n-1}) \times Area\ of\ each\ corner\ \Delta^{le}$

For More Join: @IITJEE_Advanced

$$P_1 = P_0 - 3\times\frac{1}{2}\times\left(\frac{x}{2}\right)^2\times\sin 60^\circ$$

$$P_2 = P_2 - 6\times\frac{1}{2}\times\left(\frac{x}{2^2}\right)^2\times\sin 120^\circ$$

$$P_3 = P_3 - 12\times\frac{1}{2}\times\left(\frac{x}{2^3}\right)^2\times\sin 60^\circ$$

$$(P_n) = P_{n-1} - 2^{n-1}\times 3\times\frac{1}{2}\times\left(\frac{x}{3^n}\right)^2\times\frac{\sqrt{3}}{2}$$

$$P_n = P_{n-1} - k\left(\frac{2}{9}\right)^n$$

$$P_1 = P_0 - k\left(\frac{2}{9}\right) \text{ where } \qquad k = \frac{3\sqrt{3}}{6}x^2$$

$$P_\infty = P_0 - K\left(\frac{2}{7}\right)$$

$$P_\infty = 10 - \frac{3\sqrt{3}}{8}x^2\times\frac{2}{7}$$

$$P_\infty = \frac{40}{7}$$

47. $\frac{l_1}{a_2} = \frac{l_1}{a_1} = 4 \Rightarrow l_1 = 4a_2, l_2 = 4a$

$$\frac{\frac{k}{2}(l_1 + a_1)}{\frac{k}{2}(l_2 + a_2)} = 2$$

$$4a_2 + a_1 = 2(4a_1 + a_2) = 8a_1 + 2a_2$$

$$2a_2 = 7a_1 \Rightarrow \frac{a_1}{a_2} = \frac{2}{7}$$

$$\alpha\frac{d_1}{d_2}, \lambda = \frac{l_1}{l_2} = \frac{7}{2}$$

$$\alpha = \frac{l_1 - a_1}{l_2 - a_2} = \frac{4a_2 - a_1}{4a_1 - a_2} = \frac{4a_2 - \frac{2a_2}{7}}{4\left(\frac{2a_2}{7}\right) - a_2} = \frac{\frac{4}{8} - \frac{2}{7}}{\frac{8}{7} - 1} = 26$$

48.

$$P(\overline{w}) = P(E_1 \cap \overline{w}) + P(E_2 \cap \overline{w})$$
$$= P(E_1).P(\overline{w}/E_1) + P(E_2).P(\overline{w}/E_2)$$

For More Join: @IITJEE_Advanced

$$=\frac{1}{3}.\frac{9}{10}+\frac{2}{3}.\frac{8}{10}=\frac{25}{30}=\frac{5}{6}$$

$$P(\overline{s})=P(E_1\cap\overline{s})+P(E_2\cap\overline{s})$$

$$=P(E_1).P(\overline{s}/E_1)+P(E_2).P(\overline{s}/E_2)$$

$$=\frac{1}{3}.\frac{8}{10}+\frac{2}{3}.\frac{9}{10}=\frac{26}{30}=\frac{13}{15}$$

$$P\left(\frac{E_1}{w}\right)=\frac{P(E_1).P(\overline{w}/E_1)}{P(E_1).P(\overline{w}/E_1)+P(E_2).P(\overline{w}/E_2)}$$

$$=\frac{\frac{1}{3}.\frac{9}{10}}{\frac{1}{3}.\frac{9}{10}+\frac{2}{3}.\frac{8}{10}}=\frac{9}{25}$$

49. **Sol:**

DARJEELING

4V vowels A, E, E, I

Consonants- D, R, J, L, N, G V C C V C C V C C V

$$N=6!\times\frac{4!}{2!}=2^6\times3^3\times5$$

50. Suppose that the first term of the sequence ΔA is d then

$$\Delta A=\{d,d+1,d+2,.....(d+(n-1)\}$$

Hence $A=(a_1,a_1+d,a_1+d+(d+1)$

$$a_1+d+(d+1)+(d+2).....)$$

$$a_n=a_1+(n-1)d+\frac{1}{2}(n-1)(n-2)$$

So a_n is a quadratic polynomial in n.

So $a_n=\frac{(n-\alpha)(n-\beta)}{2}$

Since $a_{19}=a_{92}=0$ we must have

$$a_n=\frac{1}{2}(n-19)(n-92)$$

So $a_3=\frac{1}{2}(3-19)(3-92)=712$

51. We can substitute $y=\sin^2 x$. Since we know that $\cos^2 x=1-\sin^2 x$, we can do some simplification.

This yields $y^5+(1-y)^5=\frac{11}{36}$. From this, we can substitute again to get some cancellation through binomials. If we let $z=\frac{1}{2}-y$, we can simplify the equation to:

$$\left(\frac{1}{2}+z\right)^5+\left(\frac{1}{2}-z\right)^5=\frac{11}{36}$$

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After using binomial theorem, this simplifies to

$$\frac{1}{16}(80z^4+40z^2+1)=\frac{11}{36}$$

If we use the quadratic formula, we obtain $z^2=\frac{1}{12}$, so $z=\pm\frac{1}{2\sqrt{3}}$ (observe that either choice of z doesn't matter). Substituting z, we get:

$$\sin^{12}x+\cos^{12}x=\left(\frac{1}{2}-z\right)^6+\left(\frac{1}{2}+z\right)^6=2z^6+\frac{15z^4}{2}+\frac{15z^2}{8}+\frac{1}{32}=\frac{13}{54}$$

Therefore, the answer is $13+54=067$.

52. we start by simplifying the first equation to

$$3\log_b\left(\sqrt{x}\log x\right)=\log_b\left(x^{\frac{3}{2}}\log^3 x\right)=56$$

$$x^{\frac{3}{2}}.\log_b^3 x=b^{56}$$

Next, we simplify the second equation to

$$\log_{\log(x)}(x)=\frac{\log_b(x)}{\log_b(\log_b(x))}=54$$

$$\log_b x=54\log_b(\log_b(x))=\log_b(\log_b^{54}(x))$$

$$x=\log_b^{54}x$$

Substituting this into the first equation gives

$$\log_b^{54-3/2}(x).\log_b^3 x=\log_b^{84}x=b^{56}$$

$$x=b^{\frac{56}{54}}=b^{\frac{2}{3}}$$

Plugging this into $x=\log_b^{54}x$ gives

$$b^{\frac{2}{3}}=\log_b^{54}(b^{\frac{2}{3}})=b^{\frac{2}{3}-54}=b^{36}$$

$$b^{\frac{2}{3}}=36$$

$$b=36^{\frac{3}{2}}=6^3=216$$

53. $$f(x)f(y)=f(x+y)+\left(\frac{\alpha}{\beta}\right)^2.xy(*)$$

In (*), we put $x=y=0\rightarrow(f(0))^2=f(0)\leftrightarrow\begin{bmatrix}f(0)=0\\f(0)=1\end{bmatrix}$

$$f(0)=0$$

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Let $x=\frac{\beta}{\alpha};y=-\frac{\beta}{\alpha}\rightarrow f\left(\frac{\beta}{\alpha}\right)f\left(-\frac{\beta}{\alpha}\right)=0\leftrightarrow\begin{bmatrix}f\left(\frac{\beta}{\alpha}\right)=0\\ f\left(-\frac{\beta}{\alpha}\right)=0\end{bmatrix}$

With $f\left(\frac{\beta}{\alpha}\right)=0$, let $y=\frac{\beta}{\alpha}\rightarrow: f\left(x+\frac{\beta}{\alpha}\right)=-\frac{\alpha}{\beta}x\leftrightarrow f(x)=-\frac{\alpha}{\beta}x+1$

With $f\left(-\frac{\beta}{\alpha}\right)=0$, let $y=-\frac{\beta}{\alpha}\rightarrow: f\left(x-\frac{\beta}{\alpha}\right)=\frac{\alpha}{\beta}x\leftrightarrow f(x)=\frac{\alpha}{\beta}x+1$

Solution: $f(x)=-\frac{\alpha}{\beta}x+1$ *or* $f(x)=\frac{\alpha}{\beta}x+1$

54. Solution:

Calculation of Standard Deviation

Class-Interval	Frequency (f)	Mid-point (m)	$d=\frac{m-750}{250}$	fd'	$f(d')^2$
0-500	90	250	-2	-180	360
500-1000	218	750	0	0	0
1000-1500	86	1250	2	172	344
1500-2000	41	1750	4	164	656
2000-3000	15	2500	7	105	735
Total	n = 450	-	-	261	2095

Here, assumed mean A is 750 and common factor C is 250.

$$S.D=C\times\sqrt{\frac{\sum fd'^2}{n}-\left(\frac{\sum fd'}{n}\right)^2}$$

$$=250\times\sqrt{\frac{2095}{450}-\left(\frac{261}{450}\right)^2}$$

$$=250\times J4.6556-(0.58)'$$

$$=250\times\sqrt{4.3192}=519.6 \text{ approximately}$$

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